

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2460889.7833 +/- 0.0022 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.23 +/- 0.028
Transit depth (Rp/Rs)^2	5.28 +/- 1.3 [%]
Orbital Inclination (inc)	85.61 +/- 1.4
Airmass coefficient 1 (a1)	0.752 +/- 0.013
Airmass coefficient 2 (a2)	0.217 +/- 0.071
Scatter in the residuals of the lightcurve fit is	1.72 %
Best Comparison Star	#3 - [532, 249]
Optimal Aperture	4.17
Optimal Annulus	12.64
Transit Duration (day)	0.048 +/- 0.03

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.23 ± 0.028 vs 0.1714 (deviation 1.46σ)
Duration fit vs published	0.048 d vs 0.0892 d (deviation 0.96σ)
Mid-time O–C	-13.6 min
Depth SNR	5.7
Residual scatter	1.72 %

Light curve

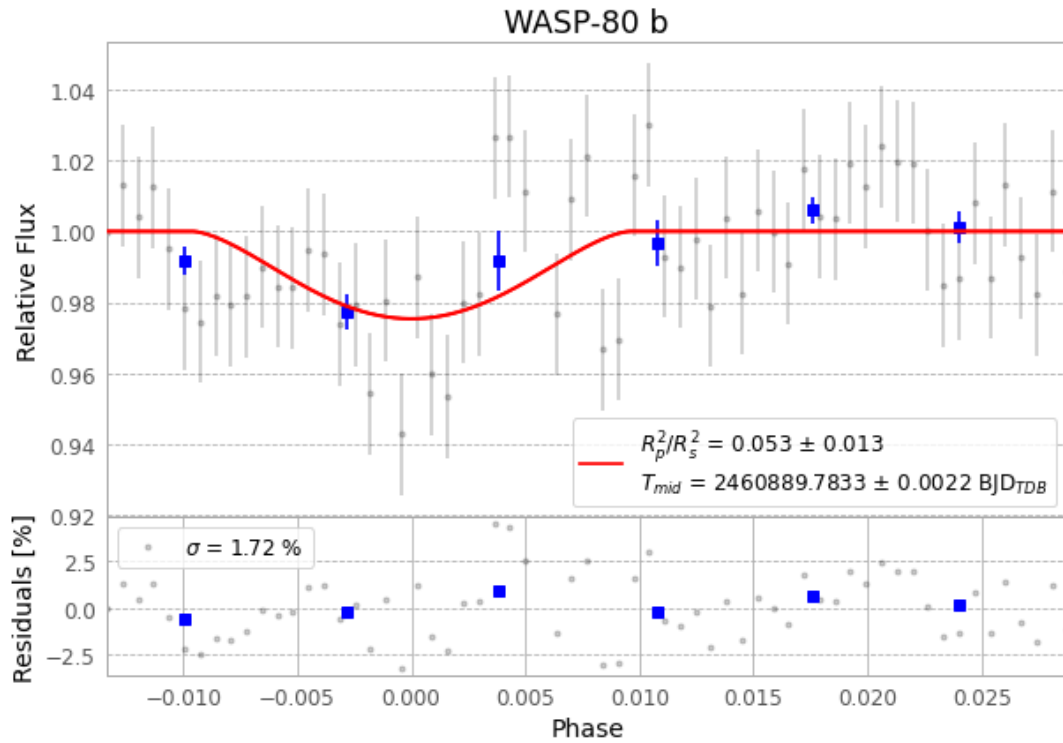


Figure 1 — FinalLightCurve_WASP-80 b_2025-08-01.png (SHA-256 24dc1a7f1b9e0ca3...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [300, 228], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d0f4e2a39dfbff04...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

63 FITS frames (41 MB), WASP-80250802053915.FITS → WASP-80250802084515.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-80-250802.log (24d7966c004e...), FinalLightCurve_WASP-80 b_2025-08-01.png (24dc1a7f1b9e...), FinalLightCurve_WASP-80 b_2025-08-01.csv (7cfc671d2486...)

Compute resources

Wall time 266.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-20 23:25Z.